

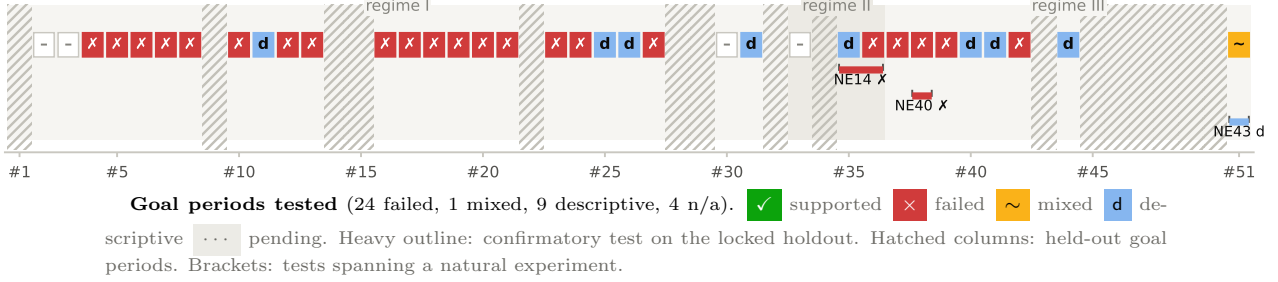
H56: Entropy production fingerprints the platform

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 1.5/5

Bottom line. Entropy production of agents' action chains does not fingerprint the platform: scaffold changes, even the regime II→III switch and an undocumented oracle swap, sit inside the day-to-day spread, which is set by the work agents do.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Does the entropy production (EP) of agents' turn-level action chains jump at scaffold changes (prompts, tools, harness) but not at goal, roster or room changes, so that EP flags undocumented platform changes? A within-agent, count-matched EP change over 3+3 days was computed at 107 dated events (changelog, NE catalog, kickoffs, joins, rooms) and at every day for blind change-point detection, on all records and on an agent-only chain with scaffold records removed. Synthetic chains came first: the estimator is calibrated and detects $\pm 50\%$ changes. On real data, scaffold changes do not stand out (0/17 tool-change hits; class $p = 0.23$), not even NE14b (-28% , $t = -1.3$) or the oracle swap (NE40), dated here to 04-20 from answer style ($t = -0.2$). Goal kickoffs move the agent-only chain more ($p = 0.007$). Family differences survive scaffold removal. Holdout not run.

PHYSICAL MODEL

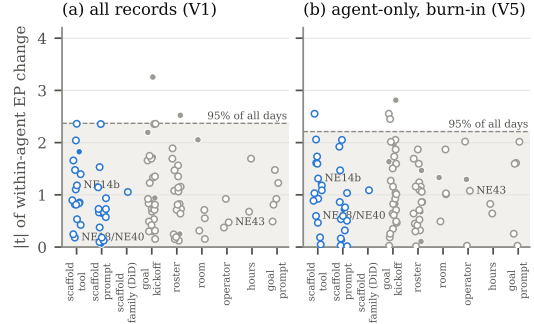
Models 02/05: each agent is a Markov chain on H14's 11 action classes (or 6 coarse states). Its EP per transition is the pair-KL between forward and reversed transitions; the AIK bound on antisymmetric transition indicators g_{ab} estimates it (Newton step, cross-fitted over days). H56 reads goal, roster and room changes as fields that move occupancy (potentials U), and scaffold changes as changes in the cycle forces F , so only the latter should change Σ .

$$\Sigma = \sum_{a,b} \pi_a P_{ab} \ln \frac{\pi_a P_{ab}}{\pi_b P_{ba}}, \quad \ln \frac{P_{ab}}{P_{ba}} = F_{ab} + U_a - U_b$$

$$\hat{\Sigma}_N = \frac{2}{k(k-1)} \sum_{f \neq f'} \bar{g}_f^\top K^{-1} \bar{g}_{f'}, \quad g_{ab} = \mathcal{K}[a \rightarrow b] - \mathcal{K}[b \rightarrow a]$$

$$\Delta_i = \hat{\Sigma}_i(\text{post}) - \hat{\Sigma}_i(\text{pre}) \quad (\text{both sides subsampled to } m_i = \min n)$$

$$t = \bar{\Delta}/\text{SE}(\Delta_i), \quad f_+ = \#\{\Delta_i > 0\}/n$$



Observables. $|t|$ of the within-agent EP change at every eligible event, by class; filled = unconfounded window, open = another event in the window; dashed = 95th percentile over all days. (a) All records. (b) Agent-only chain with a 3-transition burn-in after every scaffold cut. Scaffold changes (blue) sit inside the same band as kickoffs, joins and room changes; NE14b (regime II→III) and NE18/NE40 (the oracle swap) are ordinary.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a useful negative. An operator who hoped to watch one thermodynamic number for silent platform changes should not: behavioral EP tracks what agents are working on (goal kickoffs move it more than tool changes do), and families differ in tool-use style, not in harness. The undocumented changes we did find (the answerer swap on 04-20, a search-tool schema change on 07-29, a two-day search malfunction, the end of the operator's bookends on 08-04) all came from plain log features: event-type mix, tool-output formatting, schema fields. That is the practical monitor, and it is cheap. EP stays a descriptor of how directed agents' tool workflows are. One post hoc lead is worth a holdout check: at scaffold changes, agents' EP moves in the same direction more often than on random days.

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RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P0 synthetic calibration	false alarms 0.03–0.05; power met (fine) 0.78/0.89 at $\pm 50\%$ (fine), 0.13 (coarse)	
P1 scaffold changes jump	tool class $p = 0.23$, 0/17 hits; failed prompts 0/15	
P2 no jump: goal, roster, room	roster 0/21, room 1/5; goal 3/28 mixed ($p = 0.08$), agent-only $p = 0.007$	
P3 scaffold > goal	$\Delta t = -0.06$ ($p = 0.60$)	failed (R1)
P4 blind change-points near scaffold	$0.61\times$ chance ($p = 0.87$); no candidate with $f_+ \geq 0.8$	failed
P6 family effect vanishes	η^2 0.34 $\rightarrow$ 0.40, $p = 0.0005$ both	failed (R4)
P7 NE43 no jump	$p \geq 0.38$, 3 designs (non-diagnostic)	held
P8 NE14b jumps	8/12 agents fall, $t = -1.33$, $p = 0.33$	failed
P9 NE40 dated, EP sees it	dated 04-20 (post hoc stylometry try); $t = -0.18$	failed
P10 #51 no roster/room enrichment	$0.69\times$ ($p = 0.93$); trend $\rho = -0.40$	mixed

GOAL PERIODS IN DETAIL

G04 $\times$ failed
G05 $\times$ failed
G06 $\times$ failed
G07 $\times$ failed
G08 $\times$ failed
G10 $\times$ failed
G11 d descriptive
G12 $\times$ failed
G13 $\times$ failed
G16 $\times$ failed

+28 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

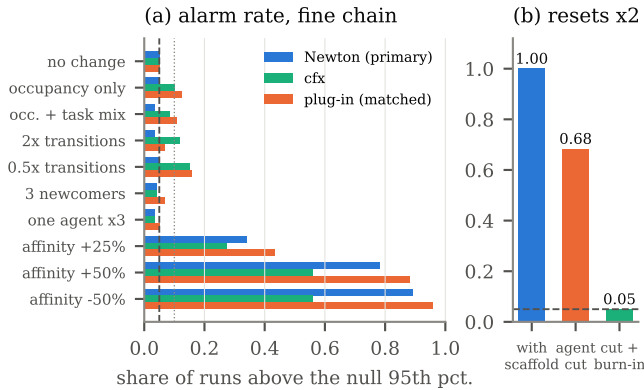
CAVEATS

Power: changes under about 30% of fine-chain EP are invisible, and coarse chains are blind; NE14b's -28% sits at the edge. The catalog is dense and confounded: only 1 tool change, 2 prompt changes and 5 kickoffs have clean windows. Changelog dates are merge dates. Providers can respond in opposite directions (NE14b), which the mean-change t dilutes. Amendment 3 (a placebo pool too small to give $p < 0.11$) was repaired after seeing data; no verdict depends on it. NE40's date came from post hoc stylometry. Values are lower bounds (non-Markov chains). Holdout not run.

ROUND 2

- **Essence**. a log-only detector of platform changes the operator did not document.
- **H56-R1** Detect platform changes from direct log features (event mix, tool schemas, output stylometry), blind-scored against the changelog.
- **H56-R2** Cross-agent sign synchrony (the post hoc lead) as the platform signature: holdout C6, per-provider DiD.
- **H56-R3** EP on Jev v3 work states, split into task-driven and scaffold-driven parts.

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Second figure. Synthetic validation at village sampling (12 agents, 3+3 days). (a) Share of runs above the null 95th percentile: the cross-fitted Newton bound ignores occupancy, count, roster and single-agent changes and detects affinity changes; the plug-in and the cross-fitted exact dual do not ignore count changes. (b) A scaffold reset leaves a footprint on the cut agent-only chain; a 3-transition burn-in removes it.